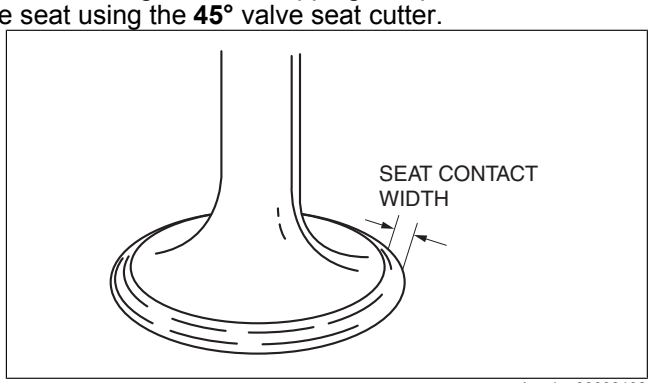
VALVE SEAT INSPECTION/REPAIR

id011000501300

1. Measure the contact width of the valve face and the valve seat using the valve lapping compound.
   - If it is not within the specification, resurface the valve seat using the **45°** valve seat cutter.

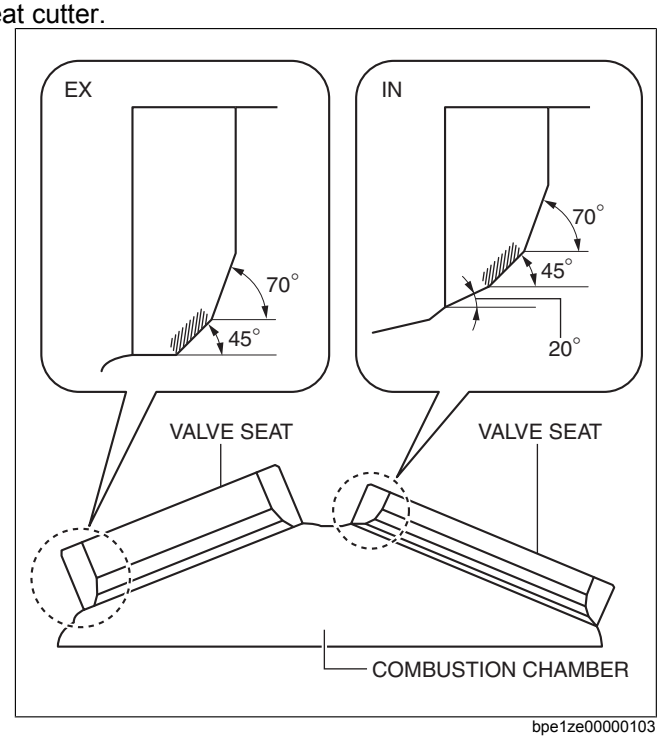
Standard valve seat contact width
1.37—1.84 mm {0.0540—0.0724 in}

Valve seat angle
45°



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2. Verify that the area where the valve seat contacts the valve face is centered.
   - If the seating position is too high, correct the valve seat using a **70°** (IN) **70°** (EX) valve seat cutter and a **45°** valve seat cutter.
   - If the seating position is too low, correct as follows:
     — IN: Correct the valve seat using a **20°** valve seat cutter and then using a **45°** valve cutter.
     — EX: Correct the valve seat using a **45°** valve seat cutter.



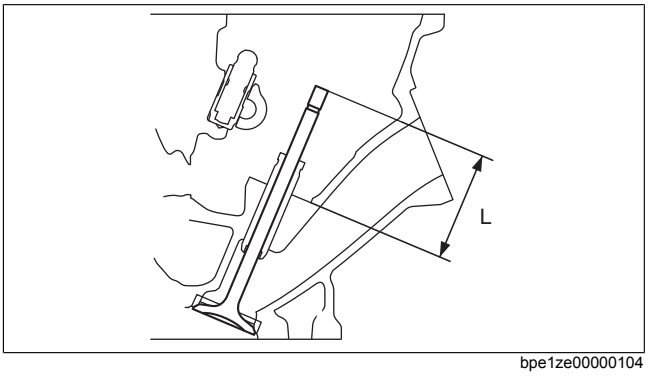
bpe1ze00000103

3. Inspect the valve seat for sinkage. Measure the protruding length (dimension L) of the valve stem using a valve of standard length.
   - If it is not within the specification, replace the cylinder head.

Standard valve seat sinkage amount (Dimension L) [SKYACTIV-G 1.3]
IN: 45.02—46.26 mm {1.773—1.821 in}
EX: 45.03—46.27 mm {1.773—1.821 in}

Standard valve seat sinkage amount (Dimension L) [SKYACTIV-G 1.5]
IN: 44.90—46.14 mm {1.768—1.816 in}
EX: 44.91—46.15 mm {1.769—1.816 in}

Standard valve seat sinkage amount (Dimension L) [SKYACTIV-G 2.0]
IN: 49.03—50.27 mm {1.931—1.979 in}
EX: 49.03—50.27 mm {1.931—1.979 in}



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Standard valve seat sinkage amount (Dimension L) [SKYACTIV-G 2.5]
IN: 48.93—50.17 mm {1.927—1.975 in}
EX: 48.87—50.11 mm {1.925—1.972 in}